



Binary Classification of Handwritten Characters





Situation



- Handwriting recognition is a very complex problem
- Full character recognition requires deep learning
- Simpler version of the problem - binary classification instead of full character recognition





Task



- Build a binary classifier
 - Use traditional ML, try to avoid deep learning to focus on core ML concepts
 - Handle real handwriting data
 - Deploy as an interactive app
- 
- A solid red vertical bar is positioned to the left of the list items.



Data Sources + Labeling



- Kaggle Datasets
 - Handwritten Letters + Digits Dataset
 - Additional digits dataset for balancing
- Folder Based Labeling
- Label rule:
 - digits → numeric (1)
 - everything else → alphabetic (0)
 - "7" → numeric
 - "A_caps" → alphabetic

> 8

> 9

> a

> A_caps

> b



Feature Engineering

- Convert images → numerical features
 - flattened pixels
 - mean & standard deviation
 - horizontal projections
 - vertical projections
- Pixel values, brightness statistics, and projections across rows and columns

A large, bold, black handwritten digit '3' is centered within a white rectangular box. The digit is thick and has a slightly irregular, hand-drawn appearance.



Models Tested/Comparison

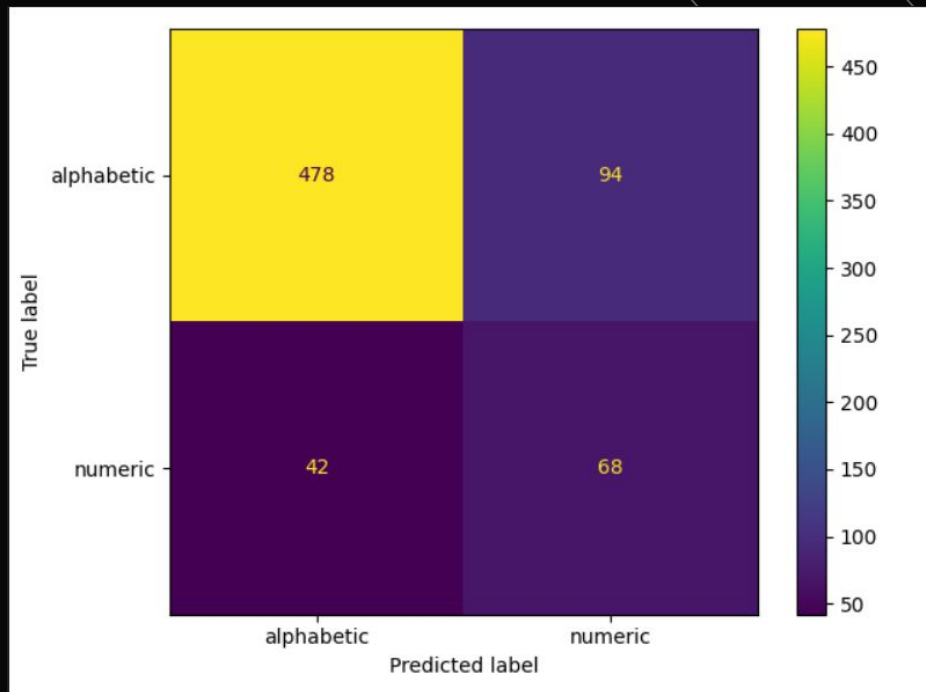


	Model	Accuracy	Numeric Recall
• Random Forest	Random Forest	~0.84	~0.45
• Logistic Regression			
• Decision Tree	KNN	~0.80	~0.57
• K Nearest Neighbors	Logistic Regression	~0.69	~0.42
	Decision Tree	~0.73	~0.45



Key Improvement

- Problem: poor numeric recall (~0.09 initially)
- Solution:
 - added second dataset
 - balanced training data
- Result: numeric recall improved (~0.60)

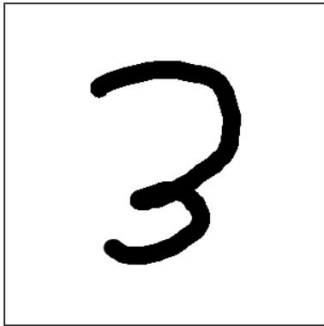


Deployment/Demo

- Flask web app, deployed with Render (doesn't work on mobile)
- User draws with mouse
- Model predicts:
 - alphabetic or numeric
 - confidence score

Draw a Letter or Number

The model will guess whether your drawing is alphabetic or numeric.



PredictClear

Result

Prediction: numeric

Confidence: 73.54%

Alphabetic probability: 26.46%

Numeric probability: 73.54%





Results/Takeaways



- Working classification system
- Feature engineering matters
- Vastly improved numeric detection
- Data balance is critical
- Interactive web app
- Different models have tradeoffs





Future Work



- Improve the model to try to predict exact characters (A vs B vs 7)
- Try CNNs
- Improve preprocessing (centering drawings)

Any Questions?

